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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, roc_curve, auc
url = r"C:\Users\shreya hajare\Downloads\Stock market.zip"
data = pd.read_csv(url)
print(data.head())
print(data.columns)
data.columns = ["Date", "Open", "High", "Low", "Close", "Shares Traded", "Turnover (? Cr)"]
data.dropna(inplace=True)
data["MA5"] = data["Close"].rolling(5).mean()
data["MA10"] = data["Close"].rolling(10).mean()
d = data["Close"].diff()
gain = d.clip(lower=0)
loss = -d.clip(upper=0)
avg_gain = gain.rolling(14).mean()
avg_loss = loss.rolling(14).mean()
rs = avg_gain / avg_loss
data["RSI"] = 100 - (100 / (1 + rs))
data["Target"] = (data["Close"].shift(-1) > data["Close"]).astype(int)
data.dropna(inplace=True)
X = data[["Open", "High", "Low", "Close", "Shares Traded", "Turnover (? Cr)"]]
scaler = StandardScaler()
X = scaler.fit_transform(X)
y = data["Target"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LogisticRegression(max_iter=3000)
model.fit(X_train, y_train)
pred = model.predict(X_test)
prob = model.predict_proba(X_test)[:, 1]
print("Accuracy:", model.score(X_test, y_test))
plt.figure(figsize=(10,5))
plt.plot(data["Close"])
plt.title("NVDA Closing Price")
plt.xlabel("Days")
plt.ylabel("Price")
plt.grid()
plt.show()
cm = confusion_matrix(y_test, pred)
plt.figure(figsize=(5,5))
plt.imshow(cm)
plt.title("Confusion Matrix")
plt.colorbar()
plt.show()
fpr, tpr, _ = roc_curve(y_test, prob)
plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label="AUC = %.2f" % auc(fpr, tpr))
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.grid()
plt.show()

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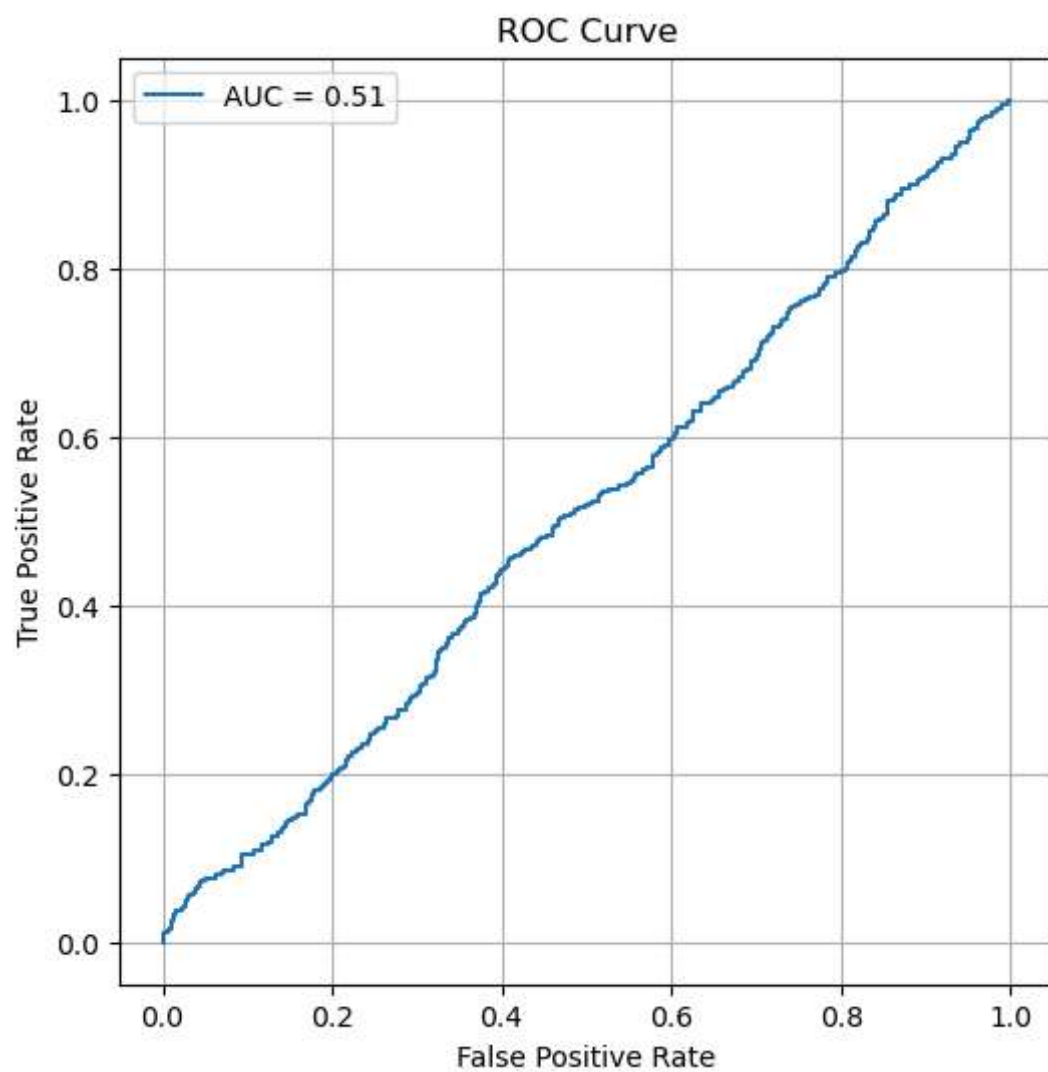
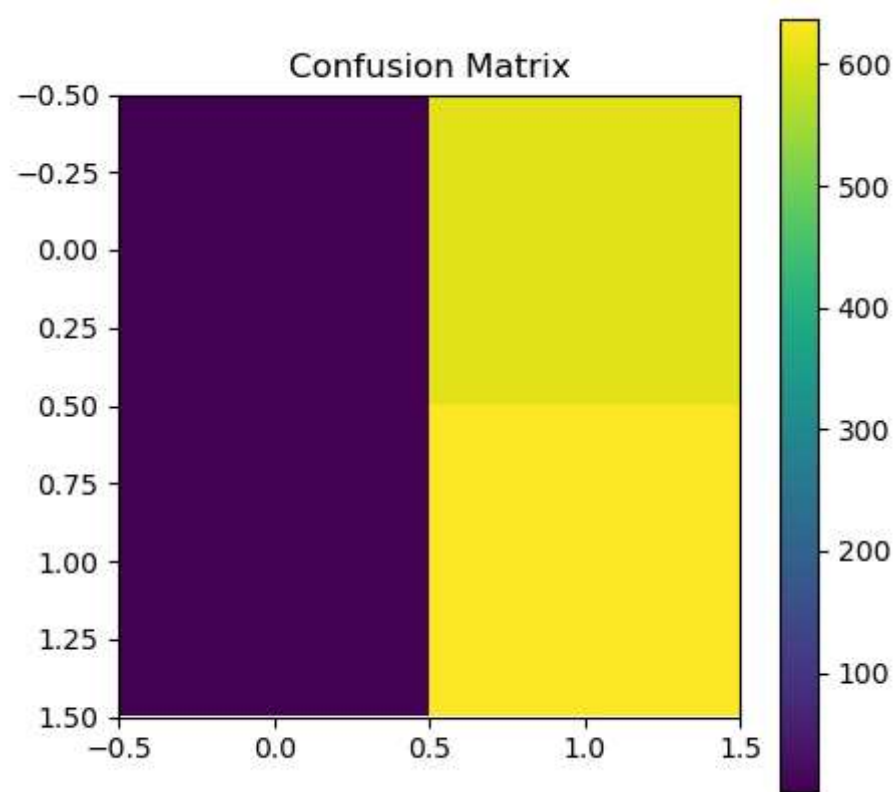
	Date	Open	High	Low	Close	Shares Traded	\
0	01 January 2001	1263.50	1276.15	1250.65	1254.30	60533274	
1	02 January 2001	1254.25	1279.60	1248.55	1271.80	72271588	
2	03 January 2001	1271.80	1293.55	1263.95	1291.25	99079153	
3	04 January 2001	1291.30	1331.35	1291.30	1307.65	106441914	
4	05 January 2001	1307.55	1330.30	1306.25	1327.25	98830568	

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Turnover (? Cr)
0      2054.04
1      2396.31
2      3065.46
3      3483.97
4      3639.04
Index(['Date ', 'Open ', 'High ', 'Low ', 'Close ', 'Shares Traded ',
      'Turnover (? Cr)'],
      dtype='object')
Accuracy: 0.5112

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In []: